

Programming SQL Server 20179

Views

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Prelude

You have learned about database, tables and constraints. You can retrieve data using SELECT query, combine data from multiple table using various types of JOIN. In this lesson we going to learn an important database object – VIEW. Views are nothing but a saved SELECT query. They do not hold data instead they retrieve data on the fly using query stored inside. Views have advantages – they allow to mask database complexities, provide security and enhance performance.

Go ahead and get started.

Today's topics

- What is view
- Creating view
- Applying option on views
- Workouts

Home works

- Add a view called Managers in the Northwind database that shows only employees that supervise other employees.
- Change the view you just created to be encrypted.

Important points

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Summary

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What is a view?

- A view is a SELECT statement stored as an Object
- A view acts like a table but does not hold data, it depends on underlying tables for data.

What can we do with a view?

- You can query a view like a table
- You can update, insert, delete data through a view [there are some restriction on inserting, updating data through a view]

Common uses for views

- Filtering rows
- Protecting sensitive data
- Reducing database complexity
- Abstracting multiple physical databases into one logical database

Advantages of views

- Mask data complexities -If you store a complex query in views., it becomes easy to query
- Provide security mechanism - Instead of giving permissions to tables, we can grant permissions to views to users and it provides a better security
- Performance enhancer - Instead of running a long SELECT statement from a client application, we can query a view. It will provide a performance gain.

Syntax of creating a view

- In simple form
CREATE VIEW view-name
AS
SELECT statement
GO
- With options
CREATE VIEW view-name
WITH
ENCRYPTION/SCHEMABINDING/VIEW_METADATA
AS
SELECT Statement
GO
- With check option
CREATE VIEW view-name
WITH
ENCRYPTION/SCHEMABINDING/VIEW_METADATA
AS
SELECT Statement
WITH CHECK OPTION
GO



The SELECT statement in a view cannot contain:

- ORDER BY clause
- COMPUTE or COMPUTE BY clause
- INTO keyword

Workout #01

- Create a table
CREATE TABLE trainees
(
 traineeid CHAR(7) primary key,
 traineeName VARCHAR(30),
 tsp CHAR(4)
)
GO
- Insert some sample data
INSERT INTO trainees VALUES
('1107320', 'Al Mamun', 'BITL'),
('1105671', 'Abdullah', 'BITL')
('1105601', 'Johora', 'HTCL'),

```
( '1125651', 'Mahmuda', 'HTCL')
```

```
GO
```

```
SELECT * FROM trainees
```

```
GO
```

- Rows in the table

traineeid	traineeName	tsp
1107320	Al Mamun	BITL
1105671	Abdullah	BITL
1105601	Johora	HTCL
1125651	Mahmuda	HTCL

- Create a view

```
CREATE VIEW vTrainees
```

```
AS
```

```
SELECT traineeid, traineeName, tsp FROM trainees
```

```
GO
```

- Test the view

```
SELECT * FROM vTrainees
```

```
GO
```

- Result

traineeid	traineeName	tsp
1107320	Al Mamun	BITL
1105671	Abdullah	BITL
1105601	Johora	HTCL
1125651	Mahmuda	HTCL

- Insert a row through the view

```
INSERT INTO vTrainees ( traineeid, traineeName, tsp)
```

```
VALUES ( '1100000', 'Test', 'ABCL')
```

```
GO
```

```
SELECT * FROM vTrainees
```

```
GO
```

- Now rows in the table

traineeid	traineeName	tsp
1107320	Al Mamun	BITL
1105671	Abdullah	BITL
1105601	Johora	HTCL
1125651	Mahmuda	HTCL
1100000	Test	ABCL

How can you extract the code of a view?

- You can use sp_helptext to see the code used to create a view
- Syntax is
EXEC sp_helptext view-name
- Try this
EXEC sp_helptext 'vTrainees'
GO
- Result should be like below
Text

```
-----  
select traineeid, traineeName, tsp from trainees
```

How can I encrypt a view?

- You can encrypt a view so that no one can extract the code used to create a view
- To do that create or alter a view with encryption option

Workout #02

- Let's alter the view we created in workout #01
- Try this
ALTER view vTrainees
WITH ENCRYPTION

```
AS
SELECT traineeid, traineeName, tsp FROM trainees
GO
```

- [Alter view follow almost the same syntax as create]
- Now try to extract text
EXEC sp_helpText vTrainees
GO
- You will see
The text for object 'vTrainees' is encrypted.

View options

- ENCRYPTION option - It protects the code you used to create a view. If you create a view with ENCRYPTION option, the code is encrypted and no one can extract the code.
- SCHEMABINDING option -If you create a view with SCHEMABINDING option, the underlying table cannot be dropped or the columns referenced by the view cannot be altered/dropped.
- VIEW_METADATA - Metadata is information about the view's properties such as column name or type. If you create a view with VIEW_METADATA option, SQL Server will use the view's metadata instead of using metadata of underlying tables.

Protecting table from alteration

- You can protect underlying tables, that a view references, from alteration
- To do that create or alter a view with SCHEMABINDING option.

Workout #03

- Try this
ALTER VIEW vTrainees
WITH ENCRYPTION, SCHEMABINDING
AS
SELECT traineeid, traineeName, tsp FROM dbo.trainees
GO
- [if you use schemabinding option, you have to use two-part table name. Listen lecture record]
- Try to alter a column
ALTER TABLE trainees
ALTER COLUMN traineeName varchar(40) NULL
GO
- Error will be like this
The object 'vTrainees' is dependent on column 'traineeName'.

Meaning of WITH CHECK option

- WITH CHECK option guarantees that the data updated or inserted through the view complies with WHERE condition used.
- It means that you cannot update a row in such a way that the row goes outside the view
- The following example clears the fact.

Workout #04

- Create a view
CREATE VIEW vTraineesInBITL
AS
SELECT * FROM trainees
WHERE tsp = 'BITL'
WITH check option
GO
- See data in view
SELECT * FROM vTraineesInBITL
GO
- Output

traineeid	traineeName	tsp
-----------	-------------	-----

1107320	Al Mamun	BITL
1105671	Abdullah	BITL

- Now try this
UPDATE vTraineesInBITL
SET tsp = 'HTCL'
WHERE traineeid='1105671'
GO
- Error

The attempted insert or update failed because the target view either specifies WITH CHECK OPTION or spans a view that specifies WITH CHECK OPTION and one or more rows resulting from the operation did not qualify under the CHECK OPTION constraint.

- Why?

Practice #01

- Create a view that selects CustomerId, CompanyName, Country, City column from Customers table in Northwind sample database
- Test the view
- Extract SQL commands used in creating the view

Practice #02

- Create a view a that return current date stripping off the time value
- Create a view that generates a random number between 10 to 20
- Create a view that shows all the user created tables in the current database

Practice #03

- Create a table Reservations as defined

Column	Data type	Nullability
Id	IDENTITY	NOT NULL
Room	INT	NOT NULL
Date	DATETIME	NOT NULL

- Insert the following data into the table

Id	Room	Date
1	105	2008-01-01
2	109	2008-02-21
3	114	2008-02-01

- Create a view that shows all the reservations in year 2008.
- Create a view that shows all the reservations in February of 2008

Practice #04

- You created a table for managing your daily expenses' information.
- You designed the table like below:

Column Name	Data Type	Nullable	Constraint
expensedate	SMALLDATETIME	-	primary key
itemName	VARCHAR(30)	no	primary key
unitprice	MONEY	no	must be > 0.0
quantity	TINYINT	no	must be >= 1

- Some sample data for the table is shown below:

expensedate	itemname	unitprice	quantity
2011-01-01	Travel bag	350.00	1
2011-01-02	Tooth paste	50.00	2
2011-01-22	Hot carrier	310.00	1
2011-02-03	Notebook	40.00	2
2011-02-10	Shopping bag	10.00	3

- A. Create a view that retrieves all data from the table.
The view adds a column amount which is calculated by multiplying unitprice and quantity
- B. Create another view that retrieves the expenses of current month
- C. Create another view that shows the sum of expenses in the current month